Lab-2

Title- Develop a Multiple Linear Regression Model using a suitable real-world open-access dataset and evaluate its performance using MSE, RMSE, and R^2 Score.

Objectives

1. To understand the concept of Multiple Linear Regression.
2. To develop a regression model using multiple independent variables.
3. To train and test the regression model using a real-world dataset.
4. To evaluate the model using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.
5. To analyze the predictive capability of the regression model.

Problem Statement

Develop a Multiple Linear Regression model using a suitable open-access real-world dataset. Train the model using multiple input features and evaluate its predictive performance using regression evaluation metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2 Score).

Theory

Multiple Linear Regression

Multiple Linear Regression is a supervised machine learning algorithm used to predict a continuous dependent variable using two or more independent variables. Unlike Simple Linear Regression, which uses only one independent variable, Multiple Linear Regression considers several features simultaneously to improve prediction accuracy.

The mathematical model is

$$Y=\beta0+\beta1X1+\beta2X2+\cdots+\beta nXn$$

Where

- $\mathbf{Y}$ = Dependent (Target) Variable
- $\mathbf{X_1, X_2,...X_n}$ = Independent Variables
- $\boldsymbol{\beta_0}$ = Intercept
- $\boldsymbol{\beta_1, \beta_2,...\beta_n}$ = Regression Coefficients

The algorithm estimates the best coefficients by minimizing the prediction error.

Regression Evaluation Metrics

1. Mean Squared Error (MSE)

MSE measures the average squared difference between actual and predicted values.

MSE=n1∑(Yi−Yi^)2

Interpretation

- Lower MSE indicates better prediction.
- Large errors are penalized more heavily.

2. Root Mean Squared Error (RMSE)

RMSE is the square root of MSE.

Interpretation

- Easier to understand because it has the same unit as the target variable.
- Lower RMSE indicates better accuracy.

3. R^2 Score (Coefficient of Determination)

R^2 measures how well the regression model explains the variation in the target variable.

Interpretation

R^2 Score	Meaning
1	Perfect prediction
0.9–1	Excellent model
0.7–0.9	Good model
0.5–0.7	Average model
<0.5	Poor model

Conclusion

Multiple Linear Regression is an effective supervised learning technique for predicting continuous values using multiple independent variables. Regression evaluation metrics such as MSE, RMSE, and R^2 Score provide quantitative measures of prediction accuracy and

model performance. Proper feature selection and pre-processing can further improve the predictive capability of the model.

Viva Questions

1. What is dimensionality reduction?
2. Why is dimensionality reduction required?
3. What is the curse of dimensionality?
4. What is Principal Component Analysis (PCA)?
5. What is Linear Discriminant Analysis (LDA)?
6. What is the difference between PCA and LDA?
7. Why is feature scaling necessary before applying PCA?
8. What are eigenvalues and eigenvectors?
9. Which technique is supervised: PCA or LDA?
10. How does dimensionality reduction improve computational efficiency?
11. What are the advantages of PCA?
12. What are the limitations of LDA?
13. Which technique is better for classification problems and why?
14. How do PCA and LDA affect model accuracy?
15. What is explained variance in PCA?